

Author Contributions Checklist

Surrogate-Assisted Local Conditional Distribution Inference with Scarce Reference Outcomes

25 August 2026

This form documents the data and code supporting the computational findings in the article and describes how to reproduce those findings.

Part 1: Data

- ☐ This paper does not involve analysis of external data.
- ☒ I certify that the authors have legitimate access to and permission to use the data analyzed in this manuscript.

Abstract

The empirical analyses use two public data sources and one author-generated synthetic benchmark. The EPA study analyzes 2,814 collocated air-sensor site-days. The VitalDB study analyzes one matched pulse-oximetry and arterial-blood-gas pair for each of 3,178 subjects. The synthetic study contains 6,000 evaluation dossiers, of which 5,997 have valid automated scores; the primary analysis uses 2,998 cases from Corpus A. Simulation outputs and derived data supporting all reported figures and tables accompany the submission.

Availability

- ☒ Data **are** publicly available.
- ☐ Data **cannot be made** publicly available.

Publicly available data

- ☒ Data are available online at:
 - EPA, *Long term performance of six air sensors across the United States*: <https://doi.org/10.23719/1531918> and <https://catalog.data.gov/dataset/dataset-long-term-performance-of-six-air-sensors-across-the-united-states>.
 - VitalDB: <https://vitaldb.net/> and its public Web API, <https://api.vitaldb.net/>. The data resource is described in <https://doi.org/10.1038/s41597-022-01411-5>.
- ☒ Data are available as part of the paper’s supplementary material.

The anonymous computational supplement contains the author-generated synthetic dossiers, expert outcomes, parsed automated-score tables, derived application data, and numerical summaries used to produce the manuscript tables and figures.
- ☐ Data are publicly available by request.
- ☐ Data are available through another mechanism.

Non-publicly available data

Not applicable to the data required to reproduce the reported findings. Raw API response caches and API credentials are not required for reproduction and are not distributed. The parsed automated scores used in the analyses are provided.

Description

File format(s)

- ☒ CSV or other plain text.

- ☒ Software-specific binary format: R data files (`.rds`).
- ☐ Standardized binary format such as netCDF or HDF5.
- ☒ Other: JSON, JSONL, and YAML files are used for synthetic dossiers, configuration, and provenance records.

Data dictionary

- ☒ Provided by the authors in the following files:
 - `data-raw/README.md`;
 - `data-raw/epa_long_term/README.md`;
 - `analysis/README.md`;
 - `reproducibility/reference_mask_split_map.md`;
 - `reproducibility/table_results_map.csv`;
 - `reproducibility/application_story/source_manifest.csv`.
- ☐ Data files are self-describing standardized binary files.
- ☐ A separate data dictionary is available at another URL.

Additional Information

The EPA source archive is retrieved by the supplied analysis code rather than redistributed. VitalDB data are retrieved through the public VitalDB interface under its access terms. The supplied synthetic dossiers contain no real-person records. The computational supplement contains no API credential.

Part 2: Code

Abstract

The supplied R package and analysis scripts implement the proposed local-CDF estimators, adaptive borrowing criterion, simulation designs, multiplier procedures, empirical analyses, and generation of manuscript tables and figures. Python code retrieves and matches the public VitalDB measurements. Shell and R entry points validate the numerical evidence and rebuild all manuscript displays. Unit tests cover the estimator, sample splitting, simulation design, application helpers, and claim-to-result correspondence.

Description

Code format(s)

- ☒ Script files
 - ☒ R
 - ☒ Python
 - ☐ Julia
 - ☐ MATLAB
 - ☐ Other
- ☒ Package
 - ☒ R
 - ☐ Python
 - ☐ Julia
 - ☐ MATLAB toolbox
 - ☐ Other
- ☐ Reproducible report
- ☒ Shell script
- ☒ Other: LaTeX manuscript and Supplement sources.

Supporting software requirements

Primary software used

- R 4.3.2;
- Python 3.12.1;

- a POSIX-compatible shell;
- TeX Live 2024 and Pandoc 3.1.12.3 for rebuilding the manuscript PDFs.

Libraries and dependencies used by the code R package versions are recorded in `renv.lock`, `reproducibility/session_info.txt`, and `reproducibility/r_package_inventory.csv`. The package-level requirements are listed in `DESCRIPTION`. Python packages used for figure production are pinned in `reproducibility/requirements-figures.txt`.

Supporting system/hardware requirements

The release was validated on Apple Silicon running macOS 15.4.1. No GPU is required. The presentation-only rebuild runs on a standard desktop. Full Monte Carlo recomputation benefits from multiple CPU cores and substantial elapsed time but does not require a computing cluster.

Parallelization used

- ☐ No parallel code used.
- ☒ Multi-core parallelization on a single machine/node.
 - Number of cores used: 4 for the manuscript-facing application and threshold-level recomputations; full simulation scripts accept a user-specified positive core count.
- ☐ Multi-machine/multi-node parallelization.

License

- ☒ MIT License.
- ☐ BSD.
- ☐ GPL v3.0.
- ☐ Creative Commons.
- ☐ Other.

Additional information

The DeepSeek scoring calls are not required for recomputation. The valid automated scores used by the manuscript are supplied, and the AI-stage scripts reuse those scores without making an external API call.

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

- ☒ Any numbers provided in text in the paper.
- ☒ The computational methods presented in the paper.
- ☒ All tables and figures in the paper.
- ☐ Selected tables and figures only.

Workflow

Location

- ☒ As part of the paper's supplementary material.
- ☐ In a Git repository disclosed during anonymous review.
- ☐ Other.

The anonymous supplementary archive is self-contained for source inspection, tests, reconstruction of manuscript-facing tables and figures, and staged recomputation. A public repository location can be added to the accepted version of this form after peer review without changing the workflow.

Format(s)

- ☒ Single master code file.
- ☒ Wrapper shell script(s).
- ☐ Self-contained Quarto, R Markdown, or Jupyter notebook.
- ☒ Text file that documents the workflow.
- ☒ Makefile.
- ☒ Other: an R package with unit tests and staged analysis scripts.

Instructions

From the root of the uncompressed anonymous supplementary archive:

1. Review `reproducibility/ANONYMOUS_REVIEW_README.md` and `renv.lock`.
2. Restore the R environment if desired with `renv::restore()` and install the Python figure dependencies from `reproducibility/requirements-figures.txt`.
3. Rebuild every main figure, the fixed-sample Supplement figure, the three main tables, and Supplement Tables S5–S6 with:

```
bash reproducibility/rebuild_submission_artifacts.sh
```

4. Run the implementation, design-correspondence, and claim-to-evidence checks with:

```
Rscript reproducibility/run_correspondence_gate.R
```

5. Inspect the complete ordered recomputation plan without changing any result file with:

```
Rscript reproduce.R 4 all
```

6. Full recomputation is staged by simulation, application, and AI components. The exact execution and finalization commands are documented in `docs/CANONICAL_RECOMPUTATION_PLAN.md`. The AI stage uses the supplied automated scores and does not call the external scoring service. After a full recomputation, `reproducibility/validate_frozen_results.R` checks the complete result store.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

- ☐ < 1 minute.
- ☐ 1-10 minutes.
- ☐ 10-60 minutes.
- ☐ 1-8 hours.
- ☒ > 8 hours.
- ☐ Not feasible to run on a desktop machine.

The display rebuild in Step 3 completes in less than one minute in the validated environment. Full Monte Carlo recomputation takes more than eight hours and is organized into independent stages. The supplied numerical results allow every reported table and figure to be rebuilt without rerunning the full Monte Carlo program.

Additional information

`README.md`, `reproducibility/README.md`, `docs/CANONICAL_RECOMPUTATION_PLAN.md`, and the package tests document the workflow and provide smaller checks of the proposed methods. The anonymous archive omits author identities, affiliations, private local paths, and API credentials; the complete unblinded archive is retained separately for the editors.